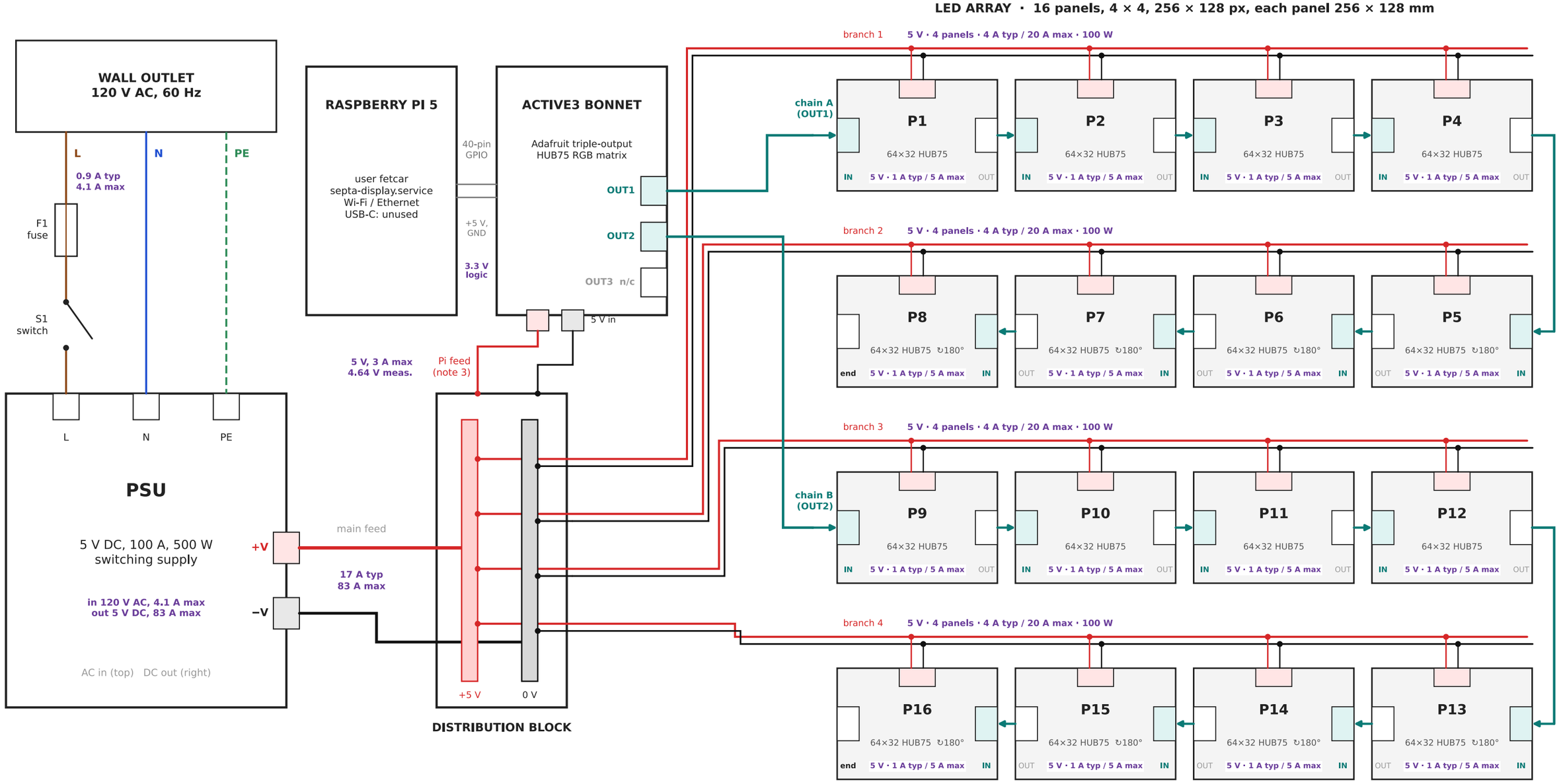


LEGEND

- +5 V DC
- 0 V / GND
- HUB75 ribbon, 5 V logic, arrow = data direction
- stacked 40-pin header
- 5 V · 20 A — rating (see table)
- mains line (L)
- mains neutral (N)
- protective earth (PE)
- junction

NOTES

1. Load: about 1 A per panel typical, 5 A at full white (E90 measurement). 16 panels → 80 A worst case, about 400 W. PSU rated 100 A. See ratings table.
2. One wall outlet. Mains passes through F1 and S1 in the PSU cord. The fuse rating was not recorded; label it when found.
3. The Pi takes 5 V from the PSU through wires soldered to the bonnet's power input, then the 40-pin header. Its USB-C input is unused. Measured 4.64 V at the Pi under full white (under-voltage limit 4.63 V). Fix: dedicated 16 AWG pair from the PSU terminals; see HARDWARE.md §1.
4. Two chains of eight on bonnet outputs 1 and 2; output 3 unused. One chain of sixteen refreshed at about 30 fps and flickered (E90 report, Methods I).
5. Serpentine: rows 2 and 4 are mounted turned 180°, so each chain doubles back. build_map() and combine_maps() in swarthmore-tracked.py encode this wiring; recabling the array needs a matching software change or the picture scrambles into blocks.
6. Panel positions are logical, as in the original sketch. Before rewiring, confirm on the hardware: which output feeds the top half (the code assumes OUT1), which corner each chain enters, whether the Pi feed leaves the block or the PSU terminals, and conductor gauges.
7. Supersedes the May 2026 hand sketch (E90 report Fig. 7), which showed a second outlet for the Pi and one 16-panel chain.

ELECTRICAL RATINGS

Circuit	Voltage	Typ	Max	Power	Wire*
Mains in, L / N	120 V AC, 60 Hz	0.9 A	4.1 A	490 W	14 AWG
PSU output (rated)	5.0 V DC		100 A	500 W	terminals
Main feed, PSU → block	5 V DC	17 A	83 A	415 W	4 AWG
Each branch, 4 panels	5 V DC	4 A	20 A	100 W	12 AWG
Each panel	5 V DC	1 A	5 A	25 W	panel lead
Pi + bonnet feed	5 V (4.64 meas.)	1 A	3 A	15 W	16 AWG
GPIO, Pi → bonnet	3.3 V logic				header
HUB75 ribbons	5 V logic				16-way IDC

typ = normal board, max = all white. Mains at 85 % PSU efficiency. Under-voltage at 4.63 V.

* Recommended minimum copper for runs under 1 m; as-built gauges not yet verified (note 6).

SEPTA SWARTHMORE STATION DISPLAY

System wiring diagram, as built

ENGR 090 E90 design project: A. Carretta, N. Fettig

Advisor: Prof. E. Masroor · Swarthmore College Engineering

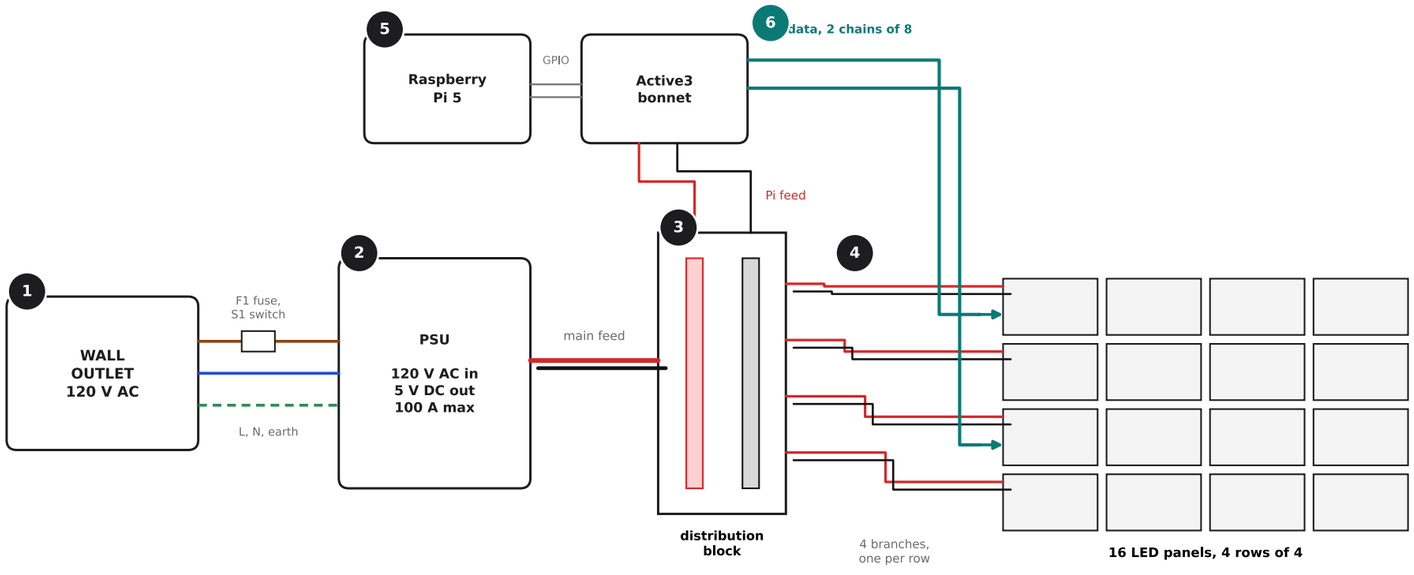
Maintainer / drawn by: B. Greenberg

github.com/BenGreenberg07/septa-tracker

DATE	REV
2026-10-21	B
SCALE	SHEET
None (schematic)	1 of 4

Two jobs: power and data

Sheet 1 shows every wire. This is the same circuit with the detail taken out. Power reaches every panel side by side (in parallel). Data goes through the panels one after another (in series). The two only share the 0 V ground.



Numbers match the walk-through on the right. Red is +5 V, black is 0 V, teal is data. Mains: brown is live, blue is neutral, green dashes are earth.

Three ideas that explain everything on these sheets

Volts, amps, watts

`W = V × A`

Volts are the push, amps are how much flows, watts are the work done. A panel at 5 V drawing 5 A uses 25 W.

Parallel adds amps

`16 × 5 A = 80 A`

Every panel sits across the same 5 V, so each gets the full voltage, and their currents add up in the wire that feeds them.

Wires lose volts

`V lost = A × Ω`

Every wire and screw has a little resistance. The more current through it, the more volts disappear before the far end. This is the Pi's problem (sheet 3).

Following the power, then the data

1. **Wall outlet, 120 V AC**
   One ordinary outlet. Live passes through fuse F1 and switch S1 on the cord. Earth is bonded to the PSU's metal case, so a loose live wire trips the breaker instead of making the case live. Draws **about 0.9 A**, **4.1 A at most**.

2. **Power supply (PSU), 5 V DC, 100 A**
   Turns mains into low-voltage DC. It holds its output at 5 V and supplies whatever current is asked for, up to 100 A. Its V-ADJ screw trims that 5 V up or down a little. Everything after this point is low voltage; everything before it is mains.

3. **Distribution block**
   Two metal bars, +5 V and 0 V. One thick main feed comes in from the PSU and five pairs go out: four branches to the panels and one to the Pi. It is a splitter, nothing more. It has no fuses.

4. **Branches and panels, 5 V, up to 5 A each**
   Each branch feeds one row of four panels in parallel. A panel takes about 1 A on an ordinary picture and up to 5 A when every LED is white, so a branch carries up to **20 A** and the whole array up to **80 A**.

5. **Raspberry Pi 5 and Active3 bonnet, 5 V, 1–3 A**
   The Pi runs the software and fetches train times. The bonnet sits on its 40-pin header and turns the Pi's **3.3 V** signals into three **5 V** HUB75 outputs. The Pi is powered backwards: its 5 V comes in through the bonnet and the header pins, not its USB-C socket.

6. **Data, two chains of eight**
   Outputs 1 and 2 each feed a chain of eight panels. Each panel passes the picture on from its OUT to the next panel's IN. Rows 2 and 4 are mounted upside down, so each chain snakes back. The software expects exactly this order.

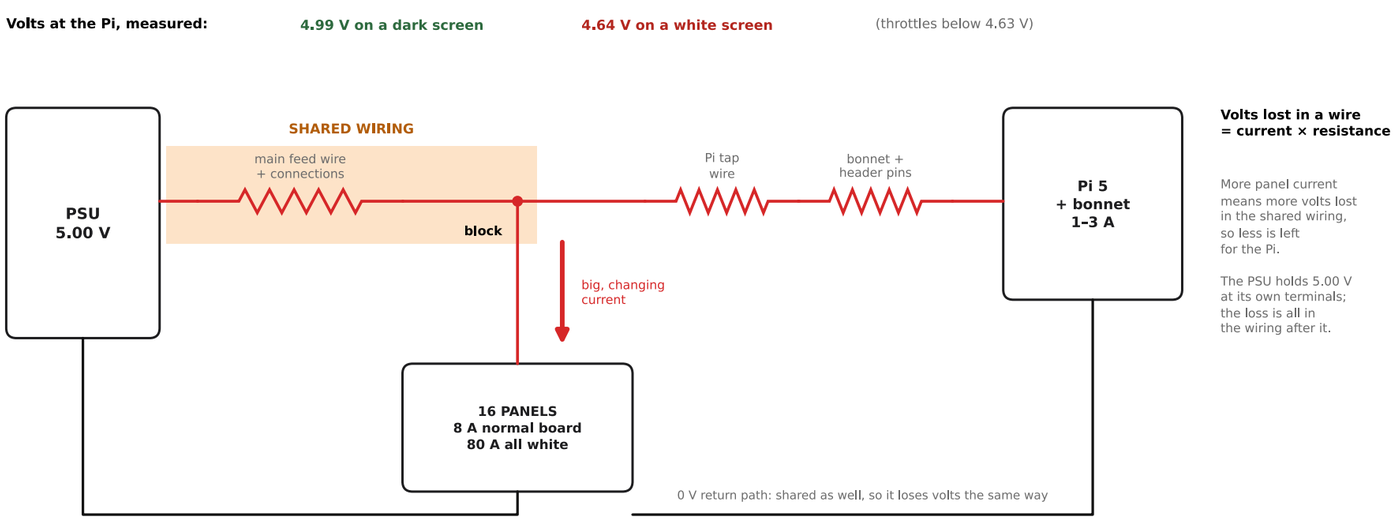
Mains stays closed

Parts 1 and 2 carry 120 V. Every change on sheet 4 is on the 5 V side. Switch off and unplug before touching anything, and work with Ed Jaoudi or Prof. Masroor present.

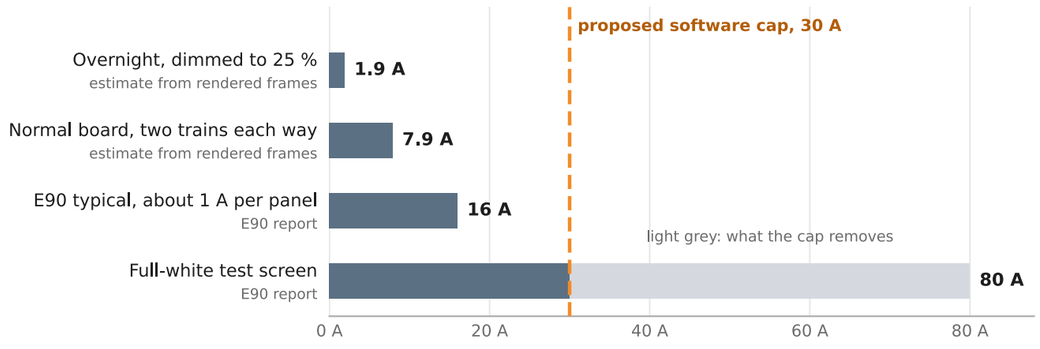
SHEET 3 · WHY THE POWER IS UNEVEN

The panels pull hard, and the Pi pays for it

The supply is big enough. The problem is where volts get lost on the way, and that nothing limits how much current the panels can ask for.



The zigzags are the resistance of wires, screws and pins, drawn as parts so you can see where volts go. The shaded run carries both loads at once.



Estimates: real board frames from the software, scaled by the E90 figure of 5 A per panel at full white. LED current only, so the true numbers are a little higher.

1 · The draw follows the picture

An LED only takes current while it is lit. A black screen takes almost nothing, and a white one lights every red, green and blue LED at once. The real board is mostly black with coloured text: about **8 A** by day and **2 A** dimmed at night. A full-white test screen takes **80 A**, ten times as much.

That swing on its own is fine. The PSU is built for it and holds 5 V at its terminals.

2 · The Pi shares wire with the panels

The Pi's feed starts at the distribution block, so the panels' current and the Pi's current both flow through the main feed and its screw connections. When the panels draw more, more volts are lost in that shared run, and the Pi gets what is left after that, its own tap wire, the bonnet and the header pins.

Measured on 2026-09-21: **4.99 V** at the Pi on a dark screen, **4.64 V** on white. The Pi throttles below **4.63 V**. It slows down there, and repeated brown-outs are how SD cards get corrupted. The flicker on white matched this exactly.

3 · Nothing limits the worst case

The software will happily show a white frame, so every wire has to be sized for 80 A even though the board never needs a fifth of that. And the block has no fuses. A 100 A supply does not trip on a pinched branch wire; it feeds it until the wire gets hot.

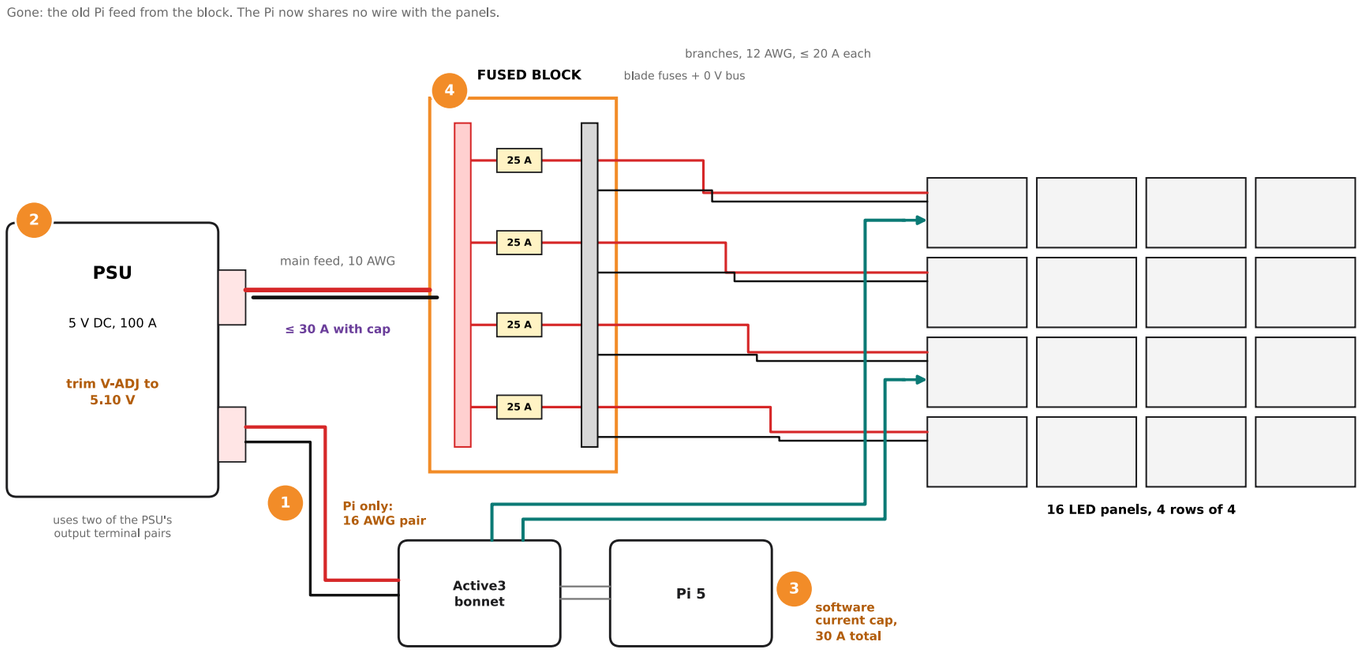
So the goal is

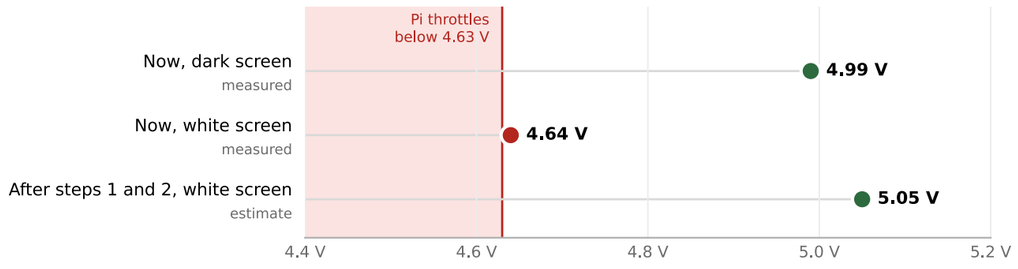
Keep the Pi's 5 V separate from the panels' current, add back a little voltage headroom, and put a ceiling on the panels' current. Then the draw is predictable and the wiring can be simple. Sheet 4 shows how.

SHEET 4 · RECOMMENDED CHANGES

Simpler, steadier power in four steps

Orange marks what changes. Do steps 1 and 2 first: together they fix the Pi. Steps 3 and 4 make the worst case predictable and safe.





Voltage at the Pi's power input (axis starts at 4.4 V)

The estimate assumes 5.10 V at the PSU, a 0.6 m 16 AWG pair (about 16 mΩ there and back) and the Pi's 3 A peak: 5.10 – 0.05 ≈ 5.05 V, whatever is on the screen.

	Do this	Why	Cost
1	**Give the Pi its own wires.** Run a red/black 16 AWG pair from a spare +V/–V screw pair on the PSU to the bonnet's 5 V input. Remove the old feed from the block.	The Pi stops sharing wire with the panels, so panel current can no longer pull its voltage down.	≈ $8
2	**Trim the PSU to 5.10 V.** Turn V-ADJ slowly while reading the PSU's output screws with a meter, board running.	A little headroom for what the wires still lose. The official Pi supply is 5.1 V too. Do not go above 5.25 V.	free
3	**Cap the current in software.** Before each frame is shown, add up its pixels; if the total would draw more than 30 A, dim that frame to fit.	The normal board (≈ 8 A) never changes. A white test screen or a bug can no longer pull 80 A, so the main feed only ever carries up to 30 A.	free
4	**Swap in a fused block.** A 6-circuit blade-fuse block with a 0 V bus (for example Blue Sea 5025) and a 25 A fuse on each branch.	A short or a pinched wire blows a $1 fuse instead of cooking the wire. The 0 V bus replaces the old ground bar.	≈ $50

Wire sizes after the change

- **Main feed:** one 10 AWG pair with the cap. Without step 3, run two 10 AWG pairs from two PSU terminal pairs.
- **Branches:** 12 AWG, up to 20 A each.
- **Pi:** 16 AWG pair, up to 3 A. Keep it short.

Check the PSU first: most 100 A units have three +V and three –V screws, so one pair for the block, one for the Pi and one spare.

How to know it worked

1. Show a white test screen. The Pi's power input reads **4.9 V or more** (it was 4.64 V).
2. `vcgencmd get_throttled` on the Pi prints `throttled=0x0` after an hour of running.
3. No flicker on white, and the panel farthest from the block reads 4.8 V or more.
4. After 10 minutes on white, no wire or screw is too hot to hold a finger on.

> **Not needed**
> A bigger PSU, a second outlet, or a separate Pi adapter. Fall back to the official 27 W USB-C supply only if step 1 does not bring the Pi to 4.9 V.